23 BCS 13189

CC

Brute force – ① Try all possible pairs A_i, A_j

② $\text{sum}(f(A_i, A_j)) \ \forall \ i, j \in [1, n]$

Code –

```
f(x, y) {
    num = x ^ y , count = 0 ;
    while (num != 0) {
        if (num % 10 == 1) count ++ ;
        num = num/10 ;
    }
    return count ;
}
main () {                                   #define MOD = 1e9+7
    int n ;
    cin>>n ;
    vector<int> arr (n) ;
    for (int i=0 ; i<n ; i++) {
        cin >> arr [i] ;
    }
    int ans = 0 ;
    for (int i=0 ; i<n ; i++) {
        for (int j=i+1 ; j<n ; j++) {
            ans += f(arr[i], arr[j]) ;
            ans = (ans % MOD) ;
        }
    }
    cout << ans ;
}
```

Time Complexity :– $O(N^2)$

Space Complexity :– $O(1)$ Exc. input array

constant

Optimised code

Approach :- ① we will count if ith bit is set or not

② for one repeat the same

③ for zero repeat the same

④ Then Using one & zero we will find the ans.

```
# include < bits /std c++ .h>
using namespace std;

int solve (vector <int> & A) {
    long long mod = 1e9 +7;
    long long n = A.size();
    long long ans = 0;
    for (int b= 0 ; b < 32 ; b++) {
        for ( int i=0 ; i<n ; i++) {
            if A[i] & (i << b)) one++;
        }
        long long zeros = n - ones;
        ans = ans + ((ones + zeroes) % MOD)* 2)) % MOD)
    }
}
```

Time Complexity :- O(n* 32)

Space Complexity :- O(1)